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PALEOMAGNETIC RESULTS FROM THE MOGOLLON-DATIL VOLCANIC FIELD, SOUTHWESTERN NEW MEXICO, AND A REFINED MID-TERTIARY REFERENCE POLE FOR NORTH AMERICA

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Abstract. Miocene-Oligocene rocks of the Mogollon-Datil volcanic field in southwestern New Mexico have yielded paleomagnetic data giving 61 site-mean/unit VGPs and a paleomagnetic pole located at 81.9°N, 143.6°E (A_{95} = 4.5°; k = 17.6). This pole is indistinguishable from other mid-Tertiary reference poles for North America. Combining the Mogollon-Datil data with VGP data from other studies of mid-Tertiary (20-40 Ma) rocks yields a paleomagnetic pole at 81.5°N, 147.3°E (N = 239; A_{95} = 2.4°; k = 15.4). When divided into normal and reversed polarity subsets the data yield paleomagnetic poles that are statistically different at the 95% confidence level [McFadden and Lowes, 1981]. The paleopole determined from reversely magnetized units, located at 78.6°N, 138.5°E (N = 125; A_{95} = 3.4°; k = 14.5), is 6.8° on the far side of the paleopole determined from normally magnetized units, positioned at 84.2°N, 166.4°E (N = 114; A_{95} = 3.3°; k = 17.0), when viewed from a common site location (i.e., the western United States). Similar pole positions an differences are also seen in 10-Ma subsets of this data set. Incomplete removal of a normal overprint on the reversely magnetized units appears unlikely to explain these results. Rather, we suggest that the differences are consistent with the concept that differing harmonics may be associated with the two polarity states over the long term [Merrill and McElhinny, 1977; Merrill et al., 1979].

Introduction

Diehl et al. [1983], in their paper establishing the Eocene and Paleocene reference poles for stable North America, also reviewed mid-Tertiary paleomagnetic poles for the North America craton. Their mean Miocene-Oligocene pole (22-38 Ma), at 83.2°N, 148.0°E, included results from the extensive mid-Tertiary ash flow sheets of Utah and Nevada [Gose, 1970; Gromme et al., 1972; Best et al., 1973; Nairn et al., 1975; Shuey et al., 1976], the Spanish Peaks dike swarm of Colorado [Larson and Strangway, 1969], the San Juan volcanic field [Tanaka and Kono, 1973; Diehl et al., 1974; Beck et al., 1977; Ellwood, 1982], the basalts from State Bridge, Colorado [York et al., 1971], and the Mistastin Lake volcanics in Canada [Currie and Larochelle, 1969]. Their pole incorporates equally weighted site-mean virtual geomagnetic poles (VGP); if the same unit was sampled by more than one study, the data were combined and a single VGP was assigned to that particular unit. This technique yielded 102 VGPs from which their reference pole was calculated. Most results used in calculating this reference pole come from rocks younger than 30 Ma. Only one study contains rocks significantly older than 30 Ma, that of the Mistastin Lakes volcanics [Currie and Larochelle, 1969]. The Miocene-Oligocene reference pole of Diehl et al. [1983] appears to be more representative of the time interval 22-30 Ma than that of 22-38 Ma.

To remove this bias in the Diehl et al. [1983] determination, a paleomagnetic study of the well-dated rocks from the Mogollon-Datil volcanic field of southwestern New Mexico was undertaken during the summers of 1981 and 1982. Simpson and Strangway [1970] and Strangway et al. [1976] were the first to obtain paleomagnetic data from these rocks. Their work concentrated on establishing a general magnetostratigraphic framework for the volcanic field. Many of the data were initially proprietary (D. W. Strangway, personal communication, 1981) and the directions were never published, although three paleomagnetic poles determined from these volcanics do appear in Appendix 4 of Strangway [1970]. Preliminary paleomagnetic results from our investigation of the Mogollon-Datil volcanic field have been reported by Diehl et al. [1982] and McClannahan et al. [1983]. McClannahan [1984] has also summarized our results from the early Oligocene units of the volcanic field. Recently, McIntosh et al. [1983, 1986] have resampled most of the major ignimbrites in the volcanic field. Using $^{40}Ar/^{39}Ar$ dating methods in conjunction with the paleomagnetic directions from these ignimbrites, they are attempting to reinterpret the stratigraphy of the Mogollon-Datil volcanic field.

Geology

The Mogollon-Datil volcanic field (Figure 1) covers an area of approximately 25,000 km^2 in southwestern New Mexico. The field is bounded by the White Mountains volcanics on the west, the Rio Grande Rift on the east, and the Colorado Plateau on the north. The volcanic field is part of the mid-Tertiary volcanic belt that stretches from Washington to Mexico. The volume of rocks present in the Mogollon-Datil field has been estimated between 50,000 and 100,000 km^3 [Elston et al., 1973; Bornhorst, 1980]. These volcanics can be subdivided into two main lithologic groups: (1) mafic lavas and breccias erupted from stratovolcanoes and (2) silicic ash flow tuffs erupted from numerous, large to small, resurgent to nonresurgent cauldrons. The eruptive centers for the two groups appear to be mutually exclusive but tend to overlap in space as well as in time. Two suites of mafic lavas and breccias have been recognized [Elston et al., 1973; Elston and Bornhorst, 1979; Bornhorst, 1980]: calc-alkalic andesites and breccias common from 42 to 32 Ma and basaltic andesites common from 32 to 18 Ma. Most of the silicic ash-flow tuffs were extruded from 34 to 25 Ma [Bornhorst, 1980].

The volcanic and tectonic evolution of the Mogollon-Datil field is similar to much of western North America during the middle to late Tertiary [Elston and Bornhorst, 1979; Elston, 1984]. It can be divided into three overlapping stages: (1) a modified Andean arc stage, (2) a

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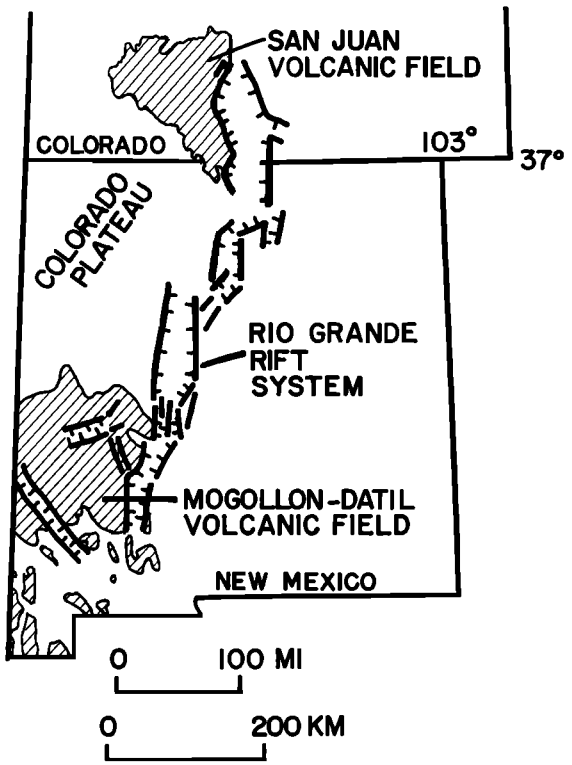


Fig. 1. Index map showing the location of the Mogollon-Datil volcanic field.

modified back arc stage, and (3) a stage of intraplate block faulting. The first stage of tectonic evolution occurred from 40 to 29 Ma and was characterized by early ductile extensional tectonics associated with subduction of the Farallon plate off the western edge of North America. In southwest New Mexico, this stage correlates with the eruption of andesites (e.g., Spears and Rubio Peak formations) and early quartz latite ash flow tuffs (e.g., Kneeling Nun, Sugarlump, Hells Mesa, Tadpole Ridge, and Bell Top tuffs) . The second stage lasted from 30 to 18 Ma and was the main stage of ductile extensional tectonics. This stage is characterized by eruption of the basaltic andesites (e.g., Bear Springs and Bearwallow Mountain basalts) and high-silica rhyolite ash flow tuffs (e.g., Bloodgood Canyon, Railroad Canyon, Davis Canyon, Vicks Peak, Shelley Peak, and La Jencia tuffs) in the Mogollon-Datil area. It correlates with the collision of the Pacific and North American plates and inception of the San Andreas transform system. The third period is Basin and Range faulting in western North America and southwest New Mexico which began about 21 Ma and is active today [Atwater, 1970; Elston and Bornhorst, 1979]. In southwestern New Mexico this period is characterized by basaltic volcanism (not studied here).

Paleomagnetic Methodology and Data

A total of 721 samples from 88 paleomagnetic sampling sites (43 in lava flow units and 45 in ash flow units) were collected from the various volcanic units of the Mogollon-Datil volcanic field. Radiometric dates of the units sampled ranged from 21-41 Ma [Bornhorst et al., 1982; Elston et al., 1973; McIntosh et al., 1986], early Miocene to late Eocene. The sampling sites were chosen to provide as wide a geographic and stratigraphic distribution as possible; however, a majority (68 sites) are from units having a radiometric age of greater than 30 Ma.

Individual cores were drilled with a portable hand-held gas-powered drill and oriented "in situ" with a Brunton compass attached to a standard orienting tool. Sun compass readings, sightings on distant known topographical features, and/or back azimuths were taken to compensate for possible orientation errors in the magnetic compass readings due to local magnetic anomalies. Attitudes were obtained by a variety of methods, depending upon the particular site in question. Methods included orientations of flattened pumices and cooling breaks within the ash flow tuffs; contacts between contrasting units, e.g., sediments versus lava flows, lava flows versus ash flow tuffs; and contacts between multiple lava flows. In many localities, more than one attitude could be obtained, and when the uncertainty in the attitude was high, the best field estimate was compared with that obtained from a geologic map by utilizing a three-point method. The majority of dips were less than or equal to 10°, the highest being 24°.

After the initial measurement of the natural remanent magnetization (NRM) vector, two pilot samples from each site were subjected to stepwise alternating field (AF) demagnetization and the results plotted on orthogonal vector endpoint diagrams. The remaining samples for the site were then demagnetized at one, two, and sometimes three AF fields in the interval over which the pilot samples displayed univectorial decay to the origin; the mean direction for the site was then determined from the demagnetization level which maximized the best estimate of the precision parameter, kappa [Fisher, 1953]. At some sites where sample behavior was not as well defined from one demagnetization level to another and required additional demagnetization steps, individual sample directions were determined by applying the principal component analysis (PCA) technique of Kirschvink [1980] to those demagnetization vectors visually defining a linear trend to the origin (Figure 2g) and the mean direction for the site calculated from these PCA directions.

Representative orthogonal vector endpoint diagrams illustrating the various types of AF demagnetization behavior typical of the Mogollon-Datil volcanic rocks are shown in Figure 2. Demagnetization to peak fields between 20 and 40 mT was sufficient to remove any secondary magnetizations, although some sites required AF fields up to 80 mT. The vector most commonly removed was the present Earth's field direction, but the vector was just as likely to be in some direction that could not be accounted for geologically. In a few instances, the linear decay to the origin of the magnetization vector observed at low AF fields (up to 40 mT) was not maintained at the higher levels of demagnetization (Figure 2g). The near complete decay of the NRM vector for most lithologies during AF demagnetization (from 1 to 40% of original NRM intensity remains at peak fields of 80-100 mT) is consistent with the contention that the remanence is carried predominantly by a titanomagnetite. Thermomagnetic results from whole rock and magnetic separates of selected units indicate further that the titanomagnetite is titanium poor (McClannahan, 1984). However, at seven sites corresponding to major ash flow sheets (Bloodgood Canyon, Davis Canyon, Shelley Peak, and the La Jencia), very little decay of the NRM vector occurred up to peak fields of 100 mT. Samples from these sites when given an isothermal remanent magnetization (IRM) did not saturate in fields up to 1 T indicating hematite contributes significantly to the remanence. Samples from these sites were subjected to thermal demagnetization up to 660 ^{o}C. As seen in Figure 2h, the thermal demagnetization continues the same univectorial decay to the origin isolated by the AF method. The mean direction determined from the thermal

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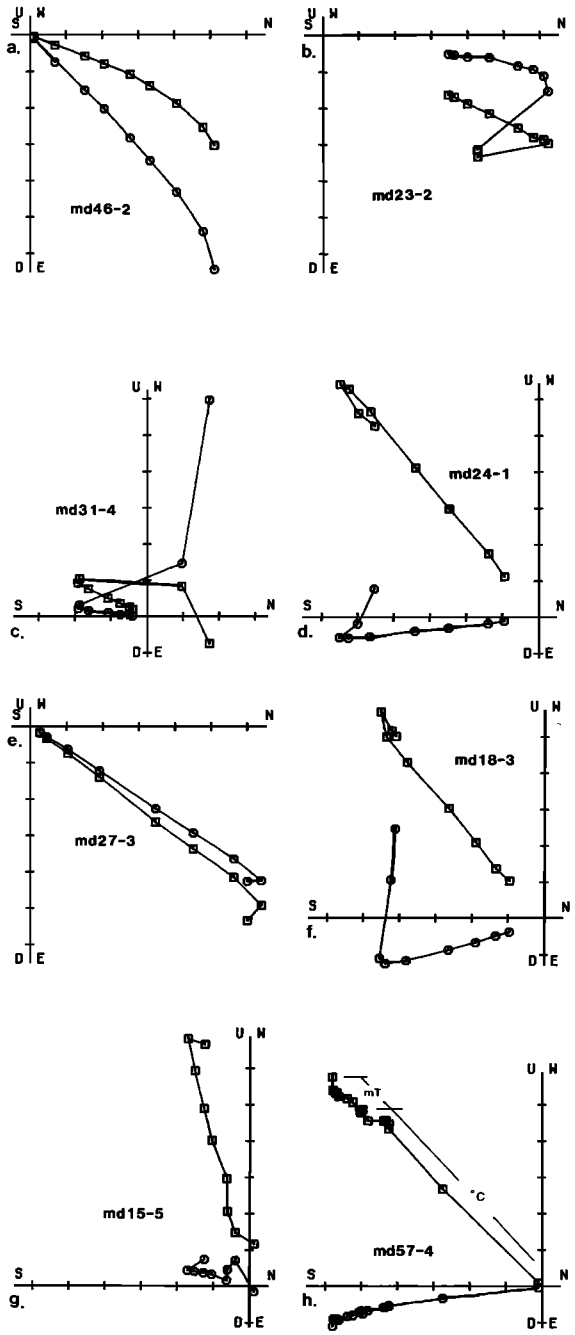


Fig. 2. Orthogonal vector endpoint demagnetization diagrams of representative samples from (a,b,c,d) lava flow and (e,f,g,h) ash flow units of the Mogollon-Datil volcanic field. Open circles are projections onto the horizontal plane; open squares are projections onto the N-S vertical plane. AF demagnetization levels are 0, 5, 10, 15, 20, 30, 40, 60, and 80 mT for all vector endpoint diagrams; thermal demagnetization levels for Figure 2f are 300, 400, 450, 500, 550, 600, 630, and 660°C. Horizontal and vertical axes divisions are (a,g) 1.8×10^{-1} A/m, (b) 1.2×10^{-1} A/m, (c) 3.6×10^{-1} A/m, (d,e) 4.8×10^{-2} A/m, (f) 5.2×10^{-2} A/m, and (h) 6.2×10^{-2} A/m.

demagnetization data in all instances was that determined from AF methods, and therefore only the AF results are presented here. McIntosh et al. [1986] have shown similar demagnetization diagrams for the units they have studied in the Mogollon-Datil volcanic field.

Table 1 summarizes the pertinent paleomagnetic data from the Mogollon-Datil sites, after AF demagnetization. Fifty-one samples from 23 sites were not used in calculating the site-mean direction/VGP for those sites. Thirty of these samples were lightning struck, and the original thermoremanent magnetization could not be resolved during AF demagnetization; all exhibited high NRM intensities and were collected from topographically high areas. Eighteen had a magnetization from which the primary or characteristic direction could not be fully recovered. The remaining three samples deviated significantly from their respective groups and were probably collected in displaced or rotated blocks.

Several sites were drilled in the same formation but at different locations. In most instances the mean directions of these sites had overlapping circles of 95% confidence, and in order not to bias the data, these sites were combined and assigned a unit weight in the final statistical analysis. Units combined in this manner were (1) the Blue Canyon Tuff, two sites; (2) the La Jencia tuff, two sites; (3) the Tadpole Ridge tuff, two sites; (4) the Datil Well tuff, two sites; (5) an unnamed ash flow tuff unit, two sites; (6) the Hells Mesa tuff, two sites; (7) the Kneeling Nun tuff, four sites; (8) the Bloodgood Canyon tuff, four sites; and (9) the Sugarlump tuff, five sites combined to form three cooling unit means.

After these sites were combined, the population size N was reduced from 88 to 74. Seven of the remaining sites were associated with alpha 95s exceeding an arbitrary cutoff value of 15° and also were not included in the final calculations. Results from six additional sites indicated VGPs more than 40° from the group mean. These data were also excluded from the final analysis, since they likely acquired their magnetization during either a reversal or excursion of the geomagnetic field or are associated with grossly inadequate tilt corrections. Similar cutoff values regarding what may be considered representative of typical secular variation have been used by McElhinny and Merrill [1975], Diehl et al. [1983], and more recently by Beck et al. [1986]. In all, these exclusions reduced the final population size to 61.

All directions/VGPs in Table 1 have been corrected for structural tilt. It was assumed that all units were deposited essentially horizontally and that any observed tilt is post magnetization. It is possible that these volcanic units were extruded over paleotopographic features and therefore the dips that are measured would in fact be original. However many of the units have dips that are correlatable to tectonic structures in the region [McClannahan, 1984], and therefore it is likely that the units were deposited prior to structural deformation and that the majority of the tilt is post eruptive in origin. Examination of the best estimate of the precision parameter, kappa, from the 61 site/unit means shows that there has been a slight improvement when the population has been corrected for tilt, from 15.1 to 18.0. Using all sites, with the exception of those whose alpha 95 exceeded 15° (i.e., N = 81), these data show a similar improvement, 11.6 to 12.4. However, neither is significant at the 95% confidence level [Cox, 1969]. In most instances, sampling was restricted to the distal portions of the ash flow sheets and lava flows where the initial dips in these units should be very low and therefore any tilt that was measured was most likely post emplacement. Since there was a slight improvement in the precision parameter of the site-mean directional data set after tectonic correction and since the mean direction before and after tilt correction differs by less than 2° (Table 2), the tilt corrected values were used instead of the uncorrected ones. Mean directions from the 61 Mogollon-Datil volcanic units after correction for tilt are shown in Figure 3.

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TABLE 1. Summary of Paleomagnetic Data From the Mogollon-Datil Volcanic Field

Site	Unit	Eruptive Type	Lat-itude	Long-itude	N/No	Ho	D	I	α_{95}	k	P_{LAT}	P_{LONG}	Age, Ma
1	Spears	lava flow	34.16	106.81	6/8	30	179.9	-36.4	10.2	43.7	76.0	73.7	33-39
2	Spears	lava flow	34.16	106.80	7/7	30	174.9	-32.0	4.0	275.4	72.6	89.7	33-39
3	Spears	ash flow	34.16	106.80	8/8	20	181.9	-34.7	2.7	431.6	74.9	66.6	33-39
4	Spears#	breccia	34.23	107.91	8/8	30	258.5	80.1	3.7	270.4	28.5	230.6	37-39
30	Spears	lava flow	34.15	106.82	7/7	20	183.7	-24.5	3.3	336.2	68.4	63.4	33-39
31	Spears	lava flow	34.15	106.82	6/7	40	178.4	-27.3	4.5	218.8	70.3	77.7	33-39
32	Spears	lava flow	34.15	106.81	8/8	40	185.4	-44.2	5.2	112.9	80.5	42.0	33-39
21	Rubio Peak#	lava flow	32.66	107.85	7/8	20	55.9	-83.6	2.8	476.6	25.1	240.7	37-41
22	Rubio Peak	lava flow	32.65	107.85	8/8	30	166.2	-57.7	2.1	710.8	77.4	193.0	37-41
23	Rubio Peak	lava flow	32.64	107.85	8/8	80	157.5	-52.9	2.0	780.4	71.2	170.9	37-41
24	Rubio Peak	lava flow	32.74	107.94	7/8	40	357.0	35.5	5.3	130.2	76.6	84.5	37-41
25	Rubio Peak	lava flow	32.74	107.94	6/8	80	20.0	42.2	12.3	30.8	70.6	2.4	37-41
62	Rubio Peak	ash flow	32.62	107.88	8/8	20	170.3	-38.3	7.3	58.4	76.0	112.4	37-41
63	Rubio Peak	lava flow	32.66	107.88	7/7	40	181.8	-40.5	3.0	421.0	80.4	62.4	37-41
64	Rubio Peak#	lava flow	32.66	107.88	7/7	60	38.8	11.3	8.4	52.9	45.0	10.4	37-41
65	Rubio Peak	lava flow	32.66	107.88	8/8	60	336.8	54.0	2.8	402.2	70.6	173.9	37-41
66	Rubio Peak	lava flow	32.64	107.87	7/7	40	166.8	-50.9	10.3	35.2	80.5	159.0	37-41
67	Rubio Peak	lava flow	32.63	107.87	5/8	pca	182.1	-51.4	8.1	89.8	88.1	0.4	37-41
74	Rubio Peak#	lava flow	32.63	107.87	6/6	40	127.0	66.5	30.2	3.6	--	--	37-41
75	Rubio Peak	lava flow	32.71	107.56	14/14	40	151.0	- 3.1	4.9	67.0	48.6	119.6	37-41
77	Rubio Peak	lava flow	32.71	107.56	7/7	60	146.9	-17.4	6.0	102.7	51.3	132.0	37-41
80	Rubio Peak#	lava flow	32.38	108.60	7/7	30	129.5	-24.3	2.4	622.9	39.9	150.4	37-41
81	Rubio Peak#	lava flow	32.38	108.60	6/7	30	120.3	-23.4	8.9	58.1	32.0	155.6	37-41
84	Rubio Peak	lava flow	32.83	107.60	7/7	20	347.2	44.0	3.0	420.4	76.9	133.4	37-41
85	Rubio Peak	lava flow	32.83	107.61	11/11	40	338.1	50.8	3.0	239.9	71.5	164.2	37-41
86	Rubio Peak	lava flow	32.83	107.61	8/8	60	340.6	62.6	4.6	142.6	71.3	204.3	37-41
87	Rubio Peak	lava flow	32.83	107.62	7/7	80	0.7	62.1	8.2	55.8	79.5	255.0	37-41
88	Rubio Peak	lava flow	32.84	107.62	8/8	40	8.2	55.9	1.9	832.5	82.3	311.8	37-41
92	Rubio Peak	lava flow	32.72	107.94	8/8	40	332.8	53.0	5.0	122.0	67.3	171.7	37-41
93	Rubio Peak	lava flow	32.72	107.94	8/8	20	334.2	58.9	6.7	69.3	68.2	187.8	37-41
5	White House Canyon	lava flow	34.18	107.92	8/8	30	4.6	70.2	2.7	428.8	69.7	259.9	37-39
6	White House Canyon	lava flow	34.16	107.85	8/8	5	17.1	67.1	3.8	215.1	59.6	285.7	37-39
7	Rock House Canyon	ash flow	34.18	107.91	8/8	40	142.3	-57.9	2.0	810.7	59.6	181.4	35
33	Hells Mesa#	ash flow	34.21	107.24	4/8	pca	170.2	-57.8	16.2	33.2	--	--	32
8	Hells Mesa	ash flow	34.08	107.91	12/12	40	144.8	-59.1	4.4	98.0	61.5	184.3	32
38	Hells Mesa	ash flow	34.13	107.87	13/13	pca	164.9	-60.5	3.1	184.1	76.0	198.0	32
8/38*					25/25		155.1	-60.2	3.1	88.1	69.2	190.0	

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TABLE 1 (continued)

Site	Unit	Eruptive Type	Lat-itude	Long-itude	N/No	Ho	D	I	α_{95}	k	P_{LAT}	P_{LONG}	Age, Ma
9	Datil Well	ash flow	34.15	107.85	7/14	pca	20.4	44.5	4.0	230.9	70.7	1.0	36
49	Datil Well	ash flow	34.08	107.85	9/9	pca	20.9	52.4	3.0	304.5	72.6	340.0	36
9/49*					16/23		20.7	48.9	2.8	168.6	72.1	349.2	
10	Unnamed	lava flow	32.99	108.22	8/8	pca	312.3	40.9	13.2	18.5	47.3	160.4	37-39
11	Unnamed	ash flow	33.90	108.21	10/10	80	136.4	-56.8	3.8	165.0	54.7	180.3	33-37
47	Unnamed	ash flow	34.16	108.01	8/8	60	132.8	-59.8	2.9	364.9	52.4	186.1	33-37
11/47*					18/18		134.9	-58.2	2.4	207.8	53.6	183.4	
12	Blue Canyon	ash flow	34.15	108.01	8/8	pca	138.3	-34.2	5.9	89.7	50.0	150.3	34
53	Blue Canyon	ash flow	33.92	108.25	7/8	pca	145.3	-22.6	3.4	312.9	51.4	135.1	34
12/53*					15/16		141.8	-28.8	4.5	71.9	51.0	143.3	
90	Tadpole Ridge	ash flow	32.89	108.24	7/8	pca	196.4	- 9.6	8.4	52.5	58.0	39.6	35
14	Tadpole Ridge	ash flow	32.89	108.24	8/8	pca	195.1	-70.2	2.6	457.6	66.2	273.9	35
15	Tadpole Ridge	ash flow	32.90	108.24	8/8	pca	204.4	-68.4	7.3	58.9	64.2	287.9	35
14/15*					16/16		199.9	-69.3	3.6	105.2	65.4	281.2	
16	Kneeling Nun	ash flow	32.77	108.11	9/9	10	166.0	-55.6	2.8	342.9	77.9	182.5	35
17	Kneeling Nun	ash flow	32.77	108.11	8/8	10	168.5	-51.5	1.5	1060.2	80.0	161.0	35
82	Kneeling Nun	ash flow	32.83	107.59	8/8	30	173.4	-50.8	3.4	271.7	84.3	151.3	35
83	Kneeling Nun	ash flow	32.83	107.60	7/7	30	163.1	-49.9	2.8	462.4	75.5	158.3	35
16/17/82/83**					4/4		167.7	-52.0	4.2	469.7	79.6	165.0	
18	Sugarlump	ash flow	32.58	107.97	7/8	30	167.8	-51.7	4.4	188.1	79.8	163.9	38
61	Sugarlump	ash flow	32.59	107.92	8/8	20	171.1	-42.5	3.3	280.1	78.8	118.6	38
18/61*					15/16		169.7	-46.8	3.4	130.0	80.1	139.0	
68	Sugarlump	ash flow	32.74	108.13	4/8	pca	344.8	70.5	7.4	156.6	65.5	230.4	38
69	Sugarlump	ash flow	32.71	108.12	7/8	pca	341.5	60.8	6.0	103.9	72.7	199.2	38
68/69*					11/16		343.9	61.8	4.2	120.1	73.5	207.4	
70	Sugarlump	ash flow	32.72	108.12	7/8	60	350.5	40.5	2.4	627.4	77.2	115.4	38
19	Faywood#	lava dome	32.56	107.97	8/8	40	168.8	44.2	60.9	1.8	--	--	33
26	Bell Top	ash flow	32.48	107.13	8/8	40	171.4	-40.6	2.6	444.8	78.0	114.4	37-40
27	Bell Top	ash flow	32.48	107.13	6/8	20	9.8	55.6	5.3	161.8	81.1	315.7	37-40
28	Bell Top	ash flow	32.48	107.13	8/8	20	336.3	71.7	1.4	1643.3	60.9	225.8	37-40
29	Bell Top	ash flow	32.48	107.13	10/10	60	347.2	56.9	5.8	69.1	78.3	192.1	37-40
34	La Jencia	ash flow	34.22	107.25	7/8	40	166.7	-50.6	4.1	217.6	78.5	151.8	29
39	La Jencia	ash flow	34.09	107.76	6/10	pca	169.9	-49.2	9.5	50.9	80.6	139.8	29
34/39*					13/18		168.2	-50.0	4.3	93.1	79.6	148.9	
35	Vicks Peak	ash flow	34.22	107.26	7/7	60	166.1	-18.4	11.2	29.7	62.1	103.2	28
36	South Crosby Peak	ash flow	34.14	107.84	8/8	pca	139.2	-43.8	4.8	132.1	53.9	159.7	32
37	South Crosby Peak	ash flow	34.12	107.88	7/8	pca	171.0	-50.5	11.4	29.2	81.9	143.8	32
40	Railroad Canyon	ash flow	33.62	108.00	7/9	80	179.0	-58.4	8.1	56.9	84.5	243.5	24
45	Railroad Canyon	ash flow	33.73	107.82	14/14	20	196.6	-50.6	1.3	1407.9	75.8	347.5	24
41	Jordan Canyon#	ash flow	33.49	108.14	10/13	40	173.3	35.7	5.3	83.4	-36.4	259.7	22
42	Double Springs	lava flow	33.47	108.13	7/7	30	3.4	47.4	3.6	290.3	84.3	40.7	23-26
43	Bearwallow Mt.	lava flow	33.58	108.09	5/5	30	176.8	-69.1	5.5	191.9	70.8	246.1	21-24

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TABLE 1 (continued)

Site	Unit	Eruptive Type	Lat-itude	Long-itude	N/No	Ho	D	I	α_{95}	k	P_{LAT}	P_{LONG}	Age, Ma
44	Bearwallow Mt.#	lava flow	33.58	108.09	5/5	pca	2.2	47.0	16.4	22.8	--	--	21-24
46	Bearwallow Mt.	lava flow	33.76	107.83	4/5	40	18.7	44.4	5.3	302.2	72.1	2.5	21-24
48	Z-Slash Canyon#	lava flow	34.06	108.06	3/9	pca	12.0	42.8	20.9	35.7	--	--	34-36
50	Squirrel Springs	lava flow	34.01	108.25	8/8	40	183.5	-23.8	3.5	247.8	68.2	62.5	29
51	Tularosa Canyon	ash flow	34.00	108.26	9/9	20	184.5	-28.1	2.5	421.7	70.4	57.6	31
52	Bloodgood Canyon	ash flow	33.73	108.33	9/9	80	164.0	-24.7	6.8	58.3	64.4	110.4	28
55	Bloodgood Canyon	ash flow	33.75	108.71	8/8	20	156.9	-33.5	1.5	1291.0	64.2	130.2	28
56	Bloodgood Canyon	ash flow	33.54	108.94	8/8	20	168.1	-26.4	3.3	289.4	67.6	102.7	28
71	Bloodgood Canyon	ash flow	33.23	108.26	10/10	20	163.1	-33.9	2.0	598.8	69.0	121.7	28
52/55/56/71**					4/4		163.2	-29.7	5.4	169.2	66.7	116.5	
57	Shelley Peak	ash flow	33.61	108.89	7/7	30	173.0	-48.7	5.7	114.0	82.8	129.3	28
59	Davis Canyon	ash flow	33.62	108.89	8/8	20	176.9	-57.0	1.9	878.1	85.3	219.3	29
72	Bear Springs	lava flow	32.72	107.86	6/6	30	21.9	75.2	4.4	231.2	57.4	271.1	27-31
91	Bear Springs	lava flow	32.85	107.87	8/8	40	166.8	-28.2	1.8	1007.2	68.5	109.0	27-31
94	Bear Springs	lava flow	32.85	107.89	7/7	20	156.9	-37.3	2.6	549.9	66.2	137.6	29
73	Mimbres Peak#	ash flow	32.68	107.84	8/8	15	80.1	2.3	29.4	2.8	--	--	33
89	Mimbres Peak	ash flow	32.84	107.63	8/8	40	341.9	36.5	7.1	61.1	69.6	129.0	33
78	Macho#	breccia	32.72	107.56	6/6	60	33.9	65.2	38.6	2.2	--	--	41
79	Macho	lava flow	32.54	107.68	7/7	30	169.2	-41.2	3.6	279.8	77.0	122.1	41

Site latitude is °N, site longitude is °W, N/No, ratio of number samples collected to those used in determining the site mean direction; Ho, optimum demagnetizing field in mT (pca indicates that three or more levels were used in determining a sample direction); D and I, mean declination and inclination for each site or unit; α_{95} and k, Fisher [1953] statistical parameters; P_{LAT} and P_{LONG}, latitude (°N) and longitude (°E) of site mean or unit VGP.

Ages based on various K-Ar (1976 IUGG decay constants), $^{40}Ar/^{39}Ar$, fission track age determinations, and stratigraphic correlations. Data from Elston et al. [1973, 1976] Bornhorst et al. [1982], Marvin and Cole, [1978], Osburn and Chapin [1983], McIntosh et al. [1986], Loring and Loring [1980], and Cather et al. [1987].

#Sites not used in calculating the position of the paleomagnetic pole.

*Individual sample directions used in calculating mean direction and VGP.

**Site-mean directions used in calculations.

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TABLE 2. Mean Direction and Paleomagnetic Pole Data From the Mogollon-Datil Volcanic Field

	Directions						Poles				
	N	D deg	I deg	α_{95}	δ	k	Lat. °N	Long. °E	A_{95}	δ	k
Before Tilt Correction											
All sites/units	61	172.6	-45.7	4.8	20.7	15.3	81.8	132.8	6.8	28.3	8.2
Normal polarity	23	351.4	53.7	6.8	17.4	20.8	82.2	185.3	7.9	20.1	15.7
Reverse polarity	38	173.2	-40.8	6.3	21.0	14.6	78.7	111.2	9.8	31.5	6.6
After Tilt Correction											
All sites/units	61	171.0	-47.3	4.4	19.0	18.1	81.9	143.6	4.5	19.2	17.6
Normal polarity	23	353.4	55.2	6.0	15.5	26.2	83.1	201.3	7.3	18.9	18.0
Reverse polarity	38	169.9	-42.4	5.7	19.2	17.4	78.7	125.4	5.4	18.2	19.5

N, number of sites/units; D, mean declination; I, mean inclination; δ, angular dispersion [Wilson, 1959]; and $\alpha(A)_{95}$ and k, statistical parameters [Fisher, 1953]. Lat. and Long., latitude and longitude.

Using the site-mean/unit-mean VGPs, we calculate a paleomagnetic pole from the Mogollon-Datil volcanic rocks at 81.9°N latitude and 143.6°E longitude (A_{95} = 4.5°; k = 17.6). Figure 4 shows the distribution of VGPs about this mean paleopole. It would appear that the rocks of the Mogollon-Datil volcanic field have adequately averaged out secular variation for the following reasons: (1) our sampling scheme covered the entire stratigraphic interval of the volcanic field (a time span of around 20 Ma), (2) the VGPs have an angular dispersion (circular standard deviation) of 19.3° with upper and lower 95% confidence limits [Cox, 1969] of 22.1° and 17.2°, respectively, which is statistically similar to the 18.4° predicted by Model F of McFadden and McElhinny [1984], and (3) the VGPs show a high degree of circularity ("a/b" = 1.10; "e" = 0.419) using the "moments of inertia" technique of Engebretson and Beck [1978] which determines the eccentricity (e) and the ratio of the major axis to minor axis (a/b) of an ellipse that best approximates the distribution. However, the mean direction from the normally magnetized units is not antiparallel to the mean direction for the reversely magnetized units at the 95% confidence level both before and after tilt correction (see Table 2). The two directions (before or after tilt correction) differ by 13.0° from being antipodal. The angular difference after correcting for tilt between the normal (D = 353.1°; I = 55.7°; α_{95} = 7.7°; N = 16) and reverse (D = 170.6°; I = -38.1°; α_{95} = 8.7°; N = 17)

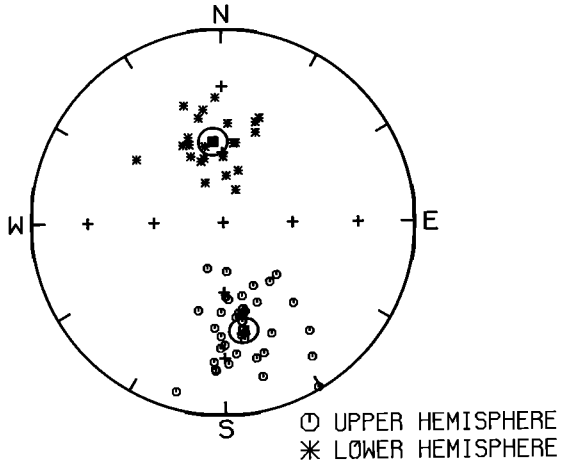


Fig. 3. Site/unit mean directions (N = 61) after tilt correction for the Mogollon-Datil volcanic rocks. Solid square is the mean of the reverse polarity sites/units (D = 169.9°; I = -42.4°; α_{95} = 5.7°), and the open square is the mean of the normal polarity sites/units (D = 353.4°; I = 55.2°; α_{95} = 6.8°). Circles about the means are the 95% confidence limits. Projection is equal area.

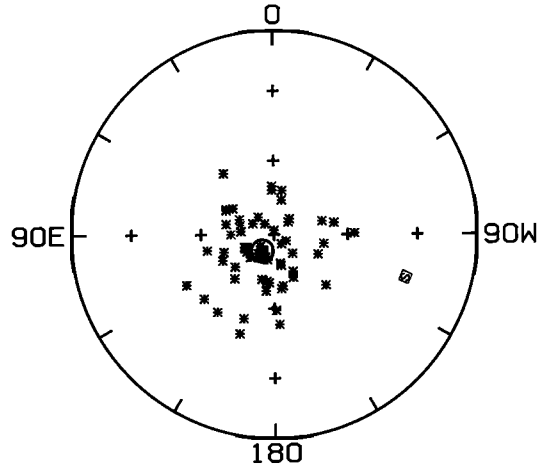


Fig. 4. Polar equal-area projection showing the distribution of VGPs from the volcanics of the Mogollon-Datil field and the mean paleomagnetic pole (square) with its circle of 95% confidence. Mean pole is located at 81.9°N, 143.6°E (A_{95} = 4.5°). [S] is sampling location.

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polarity lava flows is 17.7°, while the difference between the normal (D = 353.8°; I = 54.0°; α_{95} = 11.7°; N = 7) and reverse (D = 169.3°; I = -45.8°; α_{95} = 7.8°; N = 21) polarity ash flows is only 8.8°. In each case, the reversed polarity mean is shallower than the normal polarity mean. The possible significance of this observation is discussed below.

Last, we have not included the data of McIntosh et al. [1986] in calculating a mean paleomagnetic pole for the Mogollon-Datil volcanic field. Their study was concerned with using paleomagnetic data for stratigraphic correlation and therefore did not calculate VGP data and did not include specific site locations/structural orientations with which we could calculate VGPs. The mean direction (D = 167.5°; I = -48.4°; α_{95} = 5.6°) calculated from the data presented in Table 1 of McIntosh et al. [1986] and the mean tilt corrected direction reported in our Table 2 differ by 2.6°. Using the conditional test of McFadden and Lowes [1981], this difference is not statistically significant at the 99% confidence level. Therefore we claim that our results are representative of the Mogollon-Datil volcanic field and are not biased by omitting the McIntosh et al. [1986] data.

Discussion

The paleomagnetic pole calculated from the rocks of the Mogollon-Datil volcanic field is not statistically different from the mid-Tertiary reference pole reported by Diehl et al. [1983] in that they differ by only 1.7°. We have combined our data with those of Diehl et al. [1983] and also have added the data of Geissman et al. [1982] from the volcanic rocks of the Yerington district in Nevada, the data of Hagstrum and Lipman [1986] from the Latir volcanic field in northern New Mexico, and those of Reynolds et al. [1986] from the Lake City caldera, San Juan Mountains, Colorado. For the Latir data, magnetizations were excluded if they were considered by the authors to reflect remagnetization or if they were determined either from secondary overprints or from rocks younger than 20 Ma. For the Lake City data, fourteen sites located in the three cooling units of the Sunshine Peak tuff were combined to form six composite sites. Only the unit means were used from the Yerington study. In addition, any sites/units whose VGP was more than 40° away from the mean of this combined data set (N = 260) were eliminated from the data set before the final statistical calculations. We note that this procedure was not employed in Diehl et al.'s [1983] calculation of their mid-Tertiary reference pole.

A total of 239 site/unit VGPs from 16 different studies of rocks within the age range of 20-40 Ma provide a mid-Tertiary reference pole located at 81.5°N, 147.3°E with a circle of 95% confidence of 2.4° (Figure 5). Table 3 summarizes the pertinent paleomagnetic data. Also listed in Table 3 and shown in Figure 6 are the mean paleomagnetic poles determined from the normal and reverse polarity data subsets. These two poles differ by 6.8° of arc. This difference is statistically significant at the 95% and 99% confidence levels [McFadden and Lowes, 1981]. Furthermore, the pole from the reversed polarity data is located to the far side of the normal polarity paleopole when viewed from a common site locality (i.e., the Four Corners area of the western United States).

Two possibilities may explain the angular separation derived from the normal and reversed polarity data subsets: (1) a secondary (normal polarity) overprint has been incompletely removed from the reversely magnetized sites or (2) the data reflect that each polarity state is associated with a different non dipole harmonic content.

Incomplete demagnetization of a reversely magnetized rock which has a normal polarity/present-day overprint would cause a shallowing of the magnetization vector as discussed by Merrill and McElhinny [1983]. This is not as serious a problem for normal polarity rocks, where the direction of magnetization coincides closely with the present-day field direction, as is the situation in this study. Therefore the VGPs calculated from insufficiently cleaned reversely magnetized rocks would be farther removed from a common site locality than VGPs determined from normal polarity rocks because of the shallower inclination. We do not believe this is the cause of the discrepancy between the mean normal and reversed paleopoles listed in Table 3, as many of the investigators whose data we have used have shown that demagnetization results in a univectorial decay of the magnetization vector.

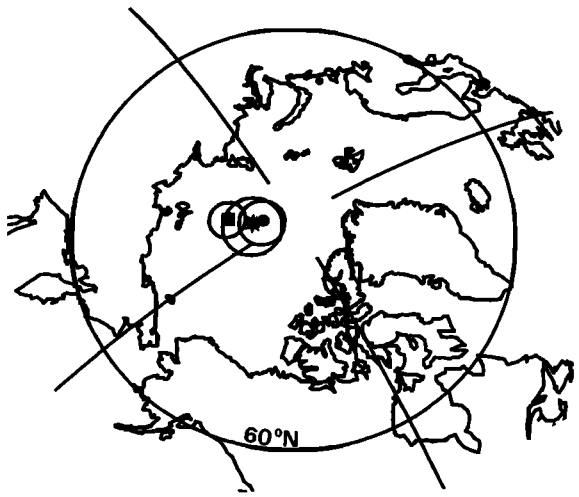


Fig. 5. Mid-Tertiary reference poles for the North American craton and their associated circles of 95% confidence. Circle, 30-Ma pole of Irving and Irving [1982]; star, 30-Ma pole of Harrison and Lindh [1982]; and square, 20- to 40-Ma reference pole of this study. Projection is oblique orthographic.

There is now well-documented evidence indicating the time-averaged paleomagnetic field contains non dipole components [Wilson and Ade-Hall, 1970; Wilson, 1970, 1971, 1972; Wilson and McElhinny, 1974; Cox, 1975; Merrill and McElhinny, 1977; Merrill et al., 1979; Coupland and Van der Voo, 1980; Livermore et al., 1983]. These long-term non dipole components, apparently largely zonal in nature, have been argued to be a significant feature of the time-averaged field for at least the past 100 m.y. These zonal components will cause measured inclinations to be shallower than predicted by the geocentric axial dipole model in the northern hemisphere. This difference between the measured inclination and the expected geocentric axial dipole direction is latitude dependent, increasing from the pole to the equator. Resulting paleomagnetic poles are therefore in error, lying on the far side of the true paleogeographic pole with respect to the sampling site locality. Besides the above inclination effect, there also appears to be a difference between normal and reverse polarity states over the long term. Wilson [1972] was the first to suggest this possibility in an analysis of paleomagnetic data from upper Tertiary and Quaternary rocks located in the USSR. Merrill and McElhinny [1977, 1983] and Merrill et al. [1979] have shown that for the last 5 m.y., VGPs determined from reversed polarity data plot more far sided than VGPs from normal polarity data, with the difference between the two means being significant. This asymmetry

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TABLE 3. Mid-Tertiary Reference Poles for North America

Ma	Polarity*	N	P_{LAT} °N	P_{LONG} °E	A_{95}	δ	k	Reference
20-40	N,R	239	81.5	147.3	2.4	20.8	15.2	This study
	N	114	84.2	166.4	3.3	19.7	17.0	This study
	R	125	78.6	138.5	3.4	21.3	14.5	This study
20-30	N,R	174	82.0	146.9	2.9	21.6	14.2	This study
	N	92	85.0	162.6	3.7	19.8	16.7	This study
	R	82	78.1	139.4	4.6	22.7	12.7	This study
30-40	N,R	65	80.5	149.4	4.2	18.5	19.0	This study
	N	22	80.6	180.6	7.1	17.8	19.9	This study
	R	43	79.5	135.7	5.1	18.4	19.2	This study
30	N,R	14#	86.0	154.0	3.0	--	144.0	Irving and Irving (1982)
30	N,R	29#	84.7	157.7	3.9	--	--	Harrison and Lindh (1982)

*N, normal; R, reversed.
#Individual study poles used.

between the two polarity states can simply be attributed to the ratio of the higher-order zonal harmonics to the axial dipole harmonic being greater for intervals of reverse polarity than for normal polarity, i.e., a "true standing field" in the terminology of Merrill et al. [1979]. Coupland and Van der Voo [1980] also have suggested that for the last 130 m.y. the magnitude of the non dipole components are greater during reversed polarity times than during normal polarity times.

Because the paleomagnetic poles presented in Table 3 represent data obtained from a latitude range less than an 8° (excluding the Mistastin Lake volcanics) and have no preferential location with respect to polarity, it is not likely that the differences between normal and reversed paleopoles are simply an artifact of an inclination effect associated with long-term zonal components of the magnetic field. Rather, we suggest these data are consistent with and further support the work of Merrill and McElhinny [1977] and Merrill et al. [1979] which indicate there are persistent differences in the time-averaged zonal harmonic terms associated with the two polarity states. This polarity asymmetry is also seen when the data presented in Table 3 are subdivided into 10-Ma intervals (20-30 and 30-40 Ma). The 10-Ma poles are nearly identical to their 20-40 Ma counterparts. The angular difference between the normal and reversed poles is significant for the 20- to 30-Ma interval (7.6°) but not significant for the 30- to 40-Ma window (6.8°) at the 95% confidence interval using the test of McFadden and Lowes [1981].

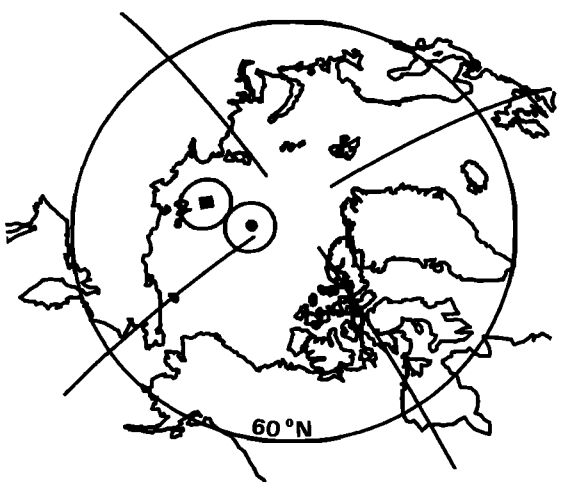


Fig. 6. Paleomagnetic poles determined from normal (circle) and reverse polarity (square) units for the interval 20-40 Ma. Projection is oblique orthographic.

Since the data only come from a small portion of the western United States, our conclusions need to be more rigorously tested using worldwide paleomagnetic data, older than 5 Ma, which are presently insufficient to test for significant differences between normal and reverse fields. For example, in a study of Miocene volcanics from southwestern Arizona [Calderone and Butler, 1984], which had experienced approximately 14° of counterclockwise rotation, the reversed polarity mean had a steeper inclination than the normal polarity mean. Therefore what we may in fact be observing is the effect of a standing or recurring non dipole feature that has biased data from this portion of the western United States.

Last, we compare our early Miocene-Oligocene reference pole with the 30-Ma reference poles of Irving and Irving [1982] and Harrison and Lindh [1982]. As seen in Figure 5, our pole is farther removed from the rotational axis than these 30-Ma reference poles, but the differences between the poles are not significant at the 95% confidence level. These observed differences most probably are a result of the method in which these poles were calculated. Both Irving and Irving [1982] and Harrison and Lindh [1982] utilized a 30-Ma "sliding" window technique. Therefore the 30-Ma pole positions they report include data from 45 to 15 Ma. Inclusion of data from rocks before 20 Ma would bias these poles toward the rotational axis. This is most easily seen utilizing the data (Table 1) of Irving and Irving [1982]. The four paleomagnetic poles between 15 and 20 Ma which were used to calculate their 30-Ma reference pole give a mean pole located at 88.6°N, 70.9°E, i.e., the rotation axis. Therefore our results provide an improved mid-Tertiary (20-40 Ma) paleomagnetic reference pole(s) for the North American craton.

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Conclusions

A paleomagnetic pole located at 81.9°N latitude, 143.6°E longitude (A_{95} = 4.5°; k = 17.6) has been calculated from the rocks of the Mogollon-Datil volcanic field. This pole is indistinguishable from other 30-Ma reference poles for the North American craton at the 95% confidence level. Combining the Mogollon-Datil data with paleomagnetic data from other mid-Tertiary volcanic units yields a paleomagnetic pole that lies at 81.5°N latitude and 147.3°E longitude (A_{95} = 2.4°; k = 15.2). However, data from only the reversely magnetized units yield a paleomagnetic pole located at 78.6°N latitude, 138.5°E longitude (A_{95} = 3.4°; k = 14.5), while data from the normally magnetized units give a paleomagnetic pole positioned at 84.2°N latitude, 166.4°E longitude (A_{95} = 3.3°; k = 17.0). These two paleopoles are separated by 6.8° of arc (significant at the 95% confidence level) with the pole determined from the reversely magnetized units located to the far side of the pole determined from the normally magnetized units when viewed from the western United States. This effect is also seen in subsets of these data sets and suggests that the difference is a real field effect and that differing harmonics may be associated with the two polarity states over the long term.

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